

Fabian Gittins

CONTACT INFORMATION	Institute for Gravitational and Subatomic Physics Princetonplein 1, Utrecht University 3584 CC Utrecht, The Netherlands	f.w.r.gittins@uu.nl fgittins.github.io +31 6 57 918 906
CITIZENSHIP	United Kingdom	
RESEARCH INTERESTS	Relativistic astrophysics, gravitational-wave astronomy and extreme physics of neutron stars. One major theme is building predictive, physically faithful neutron-star models with realistic micro-physics and dynamics. A second major theme is using gravitational waves to extract this physics and constrain dense nuclear matter. Currently, advancing gravitational-wave asteroseismology to probe dense-matter physics through modelling and detection of neutron-star oscillation modes.	
EDUCATION	PhD, Mathematics , University of Southampton, UK	Sep 2021
	Advisor: Prof Nils Andersson	
	Thesis title: <i>Gravitational waves from deformed neutron stars: mountains and tides</i>	
	MSci, Physics , University of Birmingham, UK	Jul 2017
RESEARCH EXPERIENCE	Grade: First class honours	
	Undergraduate Master's degree with focus on theoretical physics	
	Marie Skłodowska-Curie Postdoctoral Fellow , Utrecht University, NL	Oct 2024–Present
	Research Fellow , University of Southampton, UK	Oct 2021–Sep 2024
HONOURS AND AWARDS	PhD Researcher , University of Southampton, UK	Sep 2017–Sep 2021
	Marie Skłodowska-Curie Postdoctoral Fellowship (Global) , EU	Oct 2026–Sep 2029
	Project lead of <i>QuarkWave</i> ; €454,103	
	Marie Skłodowska-Curie Postdoctoral Fellowship (European) , EU	Oct 2024–Sep 2026
	Project lead of <i>DynTideEOS</i> ; €203,464	
	Gravitational Physics Thesis Prize , Institute of Physics, UK	2021
	Best Publication in Gravitational Physics , University of Southampton, UK	2021
	Physics Scholarship , University of Birmingham, UK	2013

INVITED TALKS	13. <i>Astrophysics Talks Series</i> , Istanbul University Observatory, TR (online)	2 Apr 2026
	12. <i>International Research Network for Nuclear Astrophysics Seminar</i> (online)	16 Jan 2026
	11. <i>CoCoNuT Meeting 2025</i> , University of Strasbourg, FR	28 Oct 2025
	10. <i>High Energy Particle Physics and Cosmology Theory Seminar</i> , The Johns Hopkins University, Baltimore, USA	30 Sep 2025
	9. <i>Institute for Nuclear Theory Program 25-2b</i> , University of Washington, Seattle, USA	17 Sep 2025
	8. <i>Gravitational Wave Meeting</i> , National Institute for Subatomic Physics, NL (online)	18 Jun 2025
	7. <i>Astrophysics Seminar</i> , Mullard Space Science Laboratory, University College London, UK	30 May 2024
	6. <i>Gravitational Wave Group</i> , Institute of Cosmology and Gravitation, University of Portsmouth, UK	14 Dec 2023
	5. <i>SPINS-UK Seminar</i> (online)	7 Jun 2023
	4. <i>Symposium on Gravitational Wave Physics and Astronomy: Genesis</i> , Kyoto University, JP (online)	28 Apr 2022
	3. <i>22nd BritGrav Conference</i> , University of Glasgow, UK (online)	5 Apr 2022
	2. <i>Colloquium</i> , Albert Einstein Institute, Hannover, DE (online)	6 Oct 2020
	1. <i>LIGO-Virgo Collaboration Continuous Waves Working Group</i> (online)	5 Dec 2018
CONTRIBUTED TALKS (SELECTED)	26 contributed talks at 24 separate conferences and meetings, including	
	10. <i>NNV section for (astro)particle physics fall meeting</i> , Soesterberg, NL	7 Nov 2025
	9. <i>Joint 24th International Conference on General Relativity and Gravitation and 16th Edoardo Amaldi Conference on Gravitational Waves</i> , Glasgow, UK	17 Jul 2025
	8. <i>XV Einstein Telescope Symposium</i> , Bologna, IT	27 May 2025
	7. <i>Institute for Nuclear Theory Workshop 24-89w</i> , University of Washington, Seattle, USA	5 Sep 2024
	6. <i>XIV Einstein Telescope Symposium</i> , Maastricht, NL	6–7 May 2025
	5. <i>SPINS-UK 2023 meeting</i> , Magdalen College, University of Oxford, UK	23 Nov 2023
	4. <i>Institute for Nuclear Theory Program 22-2a</i> , University of Washington, Seattle, USA	18 Jul 2022
	3. <i>PHAROS Conference 2022</i> , La Sapienza University, Rome, IT	18 May 2022
	2. <i>30th Texas Symposium on Relativistic Astrophysics</i> , University of Portsmouth, UK	17 Dec 2019
	1. <i>Joint 22nd International Conference on General Relativity and Gravitation and 13th Edoardo Amaldi Conference on Gravitational Waves</i> , Valencia, ES	9 Jul 2019

TEACHING
EXPERIENCE

Lecturer , Utrecht University, NL NS-378B, Cosmology	Apr–Jul 2026
Instructor , University of Southampton, UK MATH1007/1009, Mathematical Methods for Physical Scientists	Feb–May 2024
Guest Lecturer , University of Southampton, UK MATH3072, Advanced Fluid Dynamics MATH3006, Relativity, Black Holes and Cosmology	Oct 2022, Oct 2023 Apr 2022
Teaching Assistant , University of Southampton, UK MATH1054/1055, Mathematics for Engineering and the Environment MATH1057, Dynamics and Relativity MATH1058, Operational Research I and Mathematical Computing MATH2045, Vector Calculus and Complex Variable Theory MATH3018, Numerical Methods MATH3087, Maths and Your Future	Oct 2017–May 2021
Teaching Assistant , King Edward’s School, Birmingham, UK Physics (11–16 yr)	Jan–Apr 2016

MENTORING AND
SUPERVISION

PhD student mentoring Vincent Ernst, Université Catholique de Louvain, BE Thibaut Wouters, Utrecht University, NL Rahime Matur, University of Southampton, UK Rhys Counsell, University of Southampton, UK Shanshan Yin, University of Southampton, UK Thomas Celora, University of Southampton, UK Now postdoc at Institute of Space Sciences, Barcelona, ES	Sep 2025–Present Oct 2024–Present Jan 2023–Sep 2024 Sep 2021–Sep 2024 Sep 2021–Sep 2024 Sep 2021–Sep 2023
Bachelor’s student supervision Tobie Walraven, Utrecht University, NL	Sep 2025–Jan 2026

PROFESSIONAL ACTIVITIES, OUTREACH AND SERVICE	Virgo Collaboration, Member	Oct 2024–Present
	Cosmic Explorer Consortium, Member	May 2024–Present
	Einstein Telescope Collaboration, Member	Sep 2023–Present
	International Astronomical Union, Junior member	May 2023–Present
	International Society on General Relativity and Gravitation, Lifetime member	May 2021–Present
	European Astronomical Society, Member	Nov 2024–Dec 2025
	Royal Astronomical Society, Fellow	Jul 2021–Dec 2025
	Institute of Physics, Member	Apr 2021–Dec 2025
	Gravitational Physics Group, Committee member	Oct 2021–Sep 2025
	Conference organiser	
	SPINS-UK 2024 meeting , University of Southampton, UK Local organising committee, ~ 40 participants	10–12 Sep 2024
	Continuous gravitational waves and neutron stars workshop , Albert Einstein Institute, Hannover, DE Scientific organising committee, ~ 50 participants	17–20 Jun 2024
	Gravitational Physics Annual Meeting , Institute of Physics, UK Scientific organising committee, ~ 50 participants	18 Jan 2024
	23rd BritGrav Conference , University of Southampton, UK Scientific and local organising committee, ~ 100 participants	13–14 Apr 2023
	Seminar organiser	
	Gravity Seminar, University of Southampton, UK	Oct 2021–Sep 2024
	Weekly Gravity Reading Group, University of Southampton, UK	Jan–Jul 2021
	Journal referee	
	Astronomy and Astrophysics, Classical and Quantum Gravity, Journal of Cosmology and Astroparticle Physics, Journal of Physics G, Monthly Notices of the Royal Astronomical Society, Nature Astronomy, Physical Review D, Physical Review Letters, The Astrophysical Journal	
	Project referee	
	Postdoctoral project, University of Namur, BE	2025
	Open Fellowship, Engineering and Physical Sciences Research Council, UK	2024
	Outreach	
	Southampton Science and Engineering Festival Organised neutron-star exhibit for general public and coordinated team of 10 volunteers	7 May 2022, 18 Mar 2023
	Mathematical Challenge Marked over 200 pupil entries	Mar–Apr 2020
	Maths and Physics Workshop Demonstrated for ~ 100 secondary-school pupils	8 Nov 2017

Press (selected)

<i>Sporen van quarkmaterie in zwaartekrachtgolven?</i> Nederlands Tijdschrift voor Natuurkunde	1 Oct 2025
<i>Lightest neutron star ever found could contain compressed quarks,</i> New Scientist	24 Oct 2022
<i>Neutron star ‘mountains’ may be blocking our view of mysterious gravitational waves,</i> Live Science	21 Jul 2021
<i>Mountains on neutron stars are not even a millimetre tall due to extreme gravity,</i> The Register	21 Jul 2021
<i>Scientists find tiny mountains on neutron stars that are a fraction of a millimetre tall,</i> The Independent	19 Jul 2021
<i>Neutron Stars Have Mountains That Are Less Than a Millimeter Tall,</i> Gizmodo	18 Jul 2021
<i>Neutron stars are remarkably smooth thanks to their intense gravity,</i> New Scientist	24 May 2021
<i>Why don’t they just break up?</i> Astrobites	16 Nov 2018

COMPUTER SKILLS Advanced in Julia, Python. Intermediate in Bash, C++, Mathematica, MATLAB. Intermediate in high-performance computing (HTCondor, Slurm). Markup languages: $\text{\LaTeX}$, Markdown.

Software—Most contributions can be found at <https://github.com/fgittins>. Member of the *Bilby* development team (<https://github.com/bilby-dev/bilby>). Contributor to *SciML* (<https://sciml.ai>), in particular *NonlinearSolve.jl* (<https://github.com/SciML/NonlinearSolve.jl>). Author of *RealisticSeismology* Julia code (<https://github.com/fgittins/RealisticSeismology>).

REFERENCES

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